

Monday Warm Up: Resistors in Series/Parallel, Kirchhoff's Rules

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1 Memory Bank

- $\Delta V = IR$... Ohm's Law.
- $R_{\text{tot}} = R_1 + R_2 + \dots$... Resistors *in series*
- $R_{\text{tot}}^{-1} = R_1^{-1} + R_2^{-1} + \dots$... Resistors *in parallel*
- Kirchhoff's Rule #1: Total current into a junction equals total current out of a junction.
- Kirchhoff's Rule #2: The total change in voltage around a loop is zero.
- $P = IV = I^2R = V^2/R$... Different equations for power consumption.

2 Resistors in Series/Parallel, Kirchhoff's Rules

1. Consider Fig. 1 (top). In the circuit pictured, the resistors have values of $R_1 = 100 \, \Omega$, $R_2 = 50 \, \Omega$, and $R_3 = 50 \, \Omega$, and the battery has an emf of 120 V. (a) What is the total resistance in the circuit? (b) What is the current from the battery? (c) List the currents in each resistor.

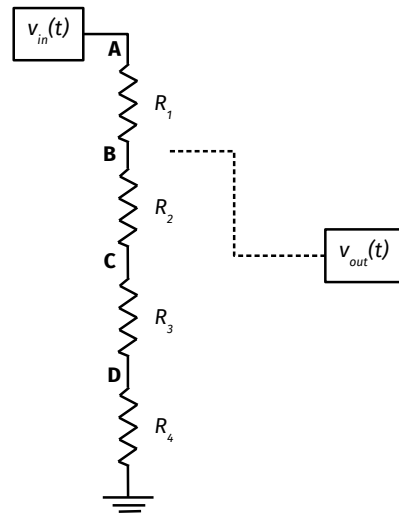
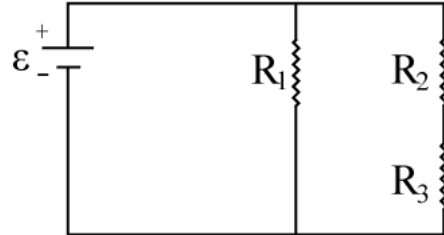


Figure 1: (Top) Three resistors connected to an EMF $\mathcal{E}$. (Bottom) A four-resistor voltage divider.

3. Consider Fig. 1 (bottom). If $v_{in} = 4.0 \, \text{V}$, what is v_{out} at points (a)-(d), if each resistor is $300 \, \Omega$?
2. What is the total power consumption of the circuit in the previous exercise?